

Pass T addendum — corrections and reconciliation

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Type: Reporting correction. No scoring change, no commits to data or config. All numbers read from docs/generated/pass_t_facts.json at commit 1dbdd46 and re-derived in-place from pass_t_facts.json.candidates[*].node_matrix.

Five items addressed. The last one changes the recommendation.

1 — ASML's tier under FR-B derived boundaries

Artifact reports: FR-B derived boundaries are `critical = 0.6357690337703887`, `high = 0.5247316525037853`, `moderate = 0.18767003183519004`. ASML's FR-B severity is `0.4667813528278457`.

Applying the lower-bound tier rule: `0.4668 < 0.5247` (below high) and `0.4668 > 0.1877` (above moderate) → **ASML tier under FR-B derived is moderate.**

Movement: committed `critical` → FR-B derived `moderate` = **2 tiers**, not 1.

Which Pass T section was right, which was wrong:

- Q9 said “ASML: *critical* → *moderate*.” **Correct.**
- The ledger said “*under FR-B + derived (critical 0.6357...) they stay high (with dysprosium's severity unchanged at 0.5618 and ASML falling to 0.4668).*” **Wrong.** ASML at 0.4668 sits below FR-B's high boundary of 0.5247; it does not stay high.

Retracted: the ledger sentence quoted above is retracted. ASML moves 2 tiers under FR-B derived, matching Q9.

2 — Reconciling the FR-B five-node tier-change list

Re-derived directly from `pass_t_facts.json.candidates["FR-B"].node_matrix`, comparing each node's committed tier to the tier its FR-B severity would sit at under FR-B's derived boundaries. Result matches `pass_t_facts.json.candidates["FR-B"].moved_under_derived` exactly:

node	committed sev	FR-B sev	committed tier	
mineral:gallium	0.4876333076	0.4876333076	high	1
mineral:dysprosium	0.5618299974	0.5618299974	critical	1
company:tsmc	0.4692819869	0.4646359779	high	1
company:asml	0.5389417592	0.4667813528	critical	1
company:applied_materials	0.2015481248	0.1647214707	moderate	1

applied_materials — does it move? Yes. Committed and FR-B side-by-side from `pass_t_facts.json`:

- Committed: `inbound_hhi = 0.0`, `outbound_criticality = 0.6766891505870234`, `severity = 0.2015481248`, `tier = moderate`, dominant axis **outbound**.
- FR-B: `inbound_hhi = 0.0`, `outbound_raw = 1.1308419667`, `outbound_normalized = 0.5530452451`, not clamped, `outbound_criticality = 0.5530452451`, `severity = 0.1647214707`, `tier_under_frozen = none`, dominant axis **outbound**.

Dominant axis before and after: outbound in both cases (inbound is 0.0, so any positive outbound wins the max). AMAT's outbound raw is 1.1308; under the frozen `fixed_reference` 1.6711 it normalizes to 0.6767; under FR-B's 2.0448 it normalizes to 0.5530. That drop of 0.1237 in concentration flows directly into severity: $0.2015 \rightarrow 0.1647$ ($\Delta = -0.0369$). Under FR-B derived's moderate boundary of 0.1877, the FR-B severity 0.1647 falls just below $\rightarrow$ none.

The Pass T ledger claim. “Net movement: 2 nodes drop under FR-B derived versus 4 under the current drift comparison.”

Directly comparable ground-truth from `pass_t_facts.json`:

- **FR-A (status quo)** moved_under_derived: 4 nodes — dysprosium (crit→high), asml (crit→high), gallium (high→mod), tsmc (high→mod). Matches the drift diagnostic in `threshold_analysis.md`.
- **FR-B** moved_under_derived: **5 nodes** — the four above plus **applied_materials (moderate→none)**, and ASML’s movement extends from 1 tier to 2 tiers.

The ledger claim was **inverted and off by a count**. Under FR-B derived boundaries, **more nodes move, not fewer**. The correct statement:

Under FR-B + derived boundaries at SF=3.0, **5 scored nodes change tier vs committed** (dysprosium, ASML, gallium, TSMC, applied_materials), with ASML moving **2 tiers down** (critical → moderate). This is more movement than the FR-A drift diagnostic shows (4 nodes, all 1-tier drops), not less. The extra movement — ASML’s second tier and AMAT’s tier drop — is a real cost of FR-B and belongs in the ship-pass attribution.

Retracted: the “Net movement: 2 nodes drop under FR-B derived versus 4 under the current drift comparison” sentence in the Pass T ledger.

3 — Blast radius for FR-C (identical shape to Q9)

Read from `pass_t_facts.json.candidates["FR-C"]`. FR-C derived boundaries are `critical = 0.5247316525037853`, `high = 0.42320867926942163`, `moderate = 0.15668443545638666`.

Nodes changing tier under FR-C + derived boundaries

`pass_t_facts.json.candidates["FR-C"].moved_under_derived` — **exactly 2 nodes**:

node	committed sev	FR-C sev	committed tier	
company:asml	0.5389417592	0.3817813806	critical	n
company:applied_materials	0.2015481248	0.1347260127	moderate	n

ASML moves the same 2 tiers under FR-C as under FR-B, but the severity path is different: under FR-C, ASML's outbound normalizes to $1.7710/2.5 = 0.7084$ (rather than FR-B's 0.8661), so the compression is larger ($\Delta = -0.1571$ vs FR-B's -0.0721). Either way, the frozen-critical membership is lost.

AMAT moves the same 1 tier (moderate $\rightarrow$ none) under both FR-B and FR-C, via the same outbound-driven mechanism — the `fixed_reference` change compresses `outbound_normalized` enough to drop the severity below the derived moderate boundary.

FR-C stability elsewhere

Under FR-C, copper, dysprosium, gallium, and TSMC all **hold their committed tier**:

- copper: sev 0.5805 > 0.5247 crit $\rightarrow$ **stays critical**
- dysprosium: sev 0.5618 > 0.5247 crit $\rightarrow$ **stays critical**
- gallium: sev 0.4876 > 0.4232 high $\rightarrow$ **stays high**
- TSMC: sev 0.4646 > 0.4232 high $\rightarrow$ **stays high**

FR-B moves all four of these; FR-C moves none of them.

Snapshot re-capture

Required. Same shape as FR-B: `severity_snapshot.json` rolls forward to a new label (e.g. `pass_p52_rebaseline_frc`), an atomic roll-forward diff (`severity_diff_pass_p52_rebaseline_frc.md`) captures the transition, and `pass_facts.py` compares against the pre-ship HEAD.

Committed artifacts regenerating

Same set as FR-B:

- docs/generated/node_inventory.md — every scored severity changes under FR-C (via the fixed_reference compression), so the inventory regenerates on every scored row.
- docs/generated/threshold_analysis.md — the drift verdict becomes “still fits the distribution” at ship time by construction; the derivation and per-boundary drift rows regenerate.
- docs/generated/severity_diff.md — the rolling diff regenerates against the new snapshot label.
- docs/generated/severity_snapshot.json — re-captured under the new label.
- docs/generated/severity_diff_pass_p52_rebaseline_frc.md — new roll-forward artifact.

Guards forecast

- backend/tests/test_thresholds_frozen.py — FROZEN_BOUNDARIES literals must be updated to FR-C’s derived triple (0.5247316525037853, 0.42320867926942163, 0.15668443545638666) in the same commit that ships FR-C. Same shape as the FR-B forecast; different literals.
- backend/tests/test_unscored.py::test_top_outbound_anchor_is_expected_node — currently asserts mineral:copper is rank-1 in raw outbound. Under FR-C, copper’s raw is still 2.0448 vs ASML’s 1.7710 (raw outbound is fixed_reference-independent) → copper is still rank-1 → **no change needed**. Same as the FR-B forecast: raw ranking is not affected by fixed_reference.
- backend/tests/test_generated_artifacts.py::test_config_boundaries_equal_derivation — currently skips under mode: frozen. Under FR-C, if the shipped literals equal the derivation output at ship time, the skip continues to be a no-op. Same as FR-B.

Pinned files

No change. known_share_offenders.txt and known_bucket_shortfalls.txt are input-share files, fixed_reference-independent. Same as FR-B forecast.

Blast-radius comparison FR-B vs FR-C

	FR-B	FR-C
tier changes under derived boundaries	5	2

	FR-B	FR-C
ASML movement	crit → moderate (2 tiers)	crit → moderate (2 tiers)
AMAT movement	moderate → none	moderate → none
dysprosium	crit → high (moves)	stays critical
gallium	high → moderate (moves)	stays high
TSMC	high → moderate (moves)	stays high
snapshot re-capture	required	required
artifacts regenerating	same set	same set
test literals updated	frozen boundaries + none else	frozen boundaries + none else
copper rank-1 test	still copper	still copper
pinned files	no change	no change

FR-C moves fewer than half as many nodes as FR-B and matches FR-B on every downstream mechanical cost.

4 — Q6/Q8 circularity: the flip test cannot fail

The user’s critique is correct.

Q6 framed the K.1-warning question as: *“will you re-anchor again the next time the graph max moves?”* Answering “only under another authorizing spec” was the ground for calling FR-B pattern-not-syntax.

Q8 proposed as the flip-to-FR-C condition: *“if the drift diagnostic under Pass T close still reports 4 would-change-tier under derived boundaries after FR-B is committed.”*

But drift is defined as *frozen boundaries vs derivation from current distribution*. If FR-B ships with frozen boundaries set to FR-B’s derived boundaries — which is what the recommendation is — then at ship time, frozen = derived by construction and drift reports **o would-change-tier**. The flip test cannot return the refuting answer. This is the K.2.2 §6.2 pattern: a measurement whose outcome is entailed by the way it is set up.

Instance count of this pattern in project ledger:

1. K.1 §6.2
2. K.2 §6.3
3. K.2.1 §6.2
4. Pass S retraction (asserting max-path from a null result — retracted in the Pass S report correction commit 32a1ff4)
5. **Pass T Q8 flip condition — this one.**

Now the fifth instance.

Replacement flip condition (adopted).

After FR-B ships, does any future pass’s authoring push a node’s raw outbound above 2.0447548854281186?

Falsifiability check:

- **Can return yes.** A future re-authoring pass could raise the `input_share` on any edge in a chain leading to a high-fan-out destination. Specifically, TSMC’s raw (currently 1.7524) is a plausible candidate: if a future pass re-authors edges into TSMC’s downstream customers (e.g., new cowos → new-consumer edges, or new supplies → new-fab edges), TSMC’s max-of-paths outbound can rise. Similarly `ndfeb_magnets` → facility edges, if re-authored on §4 basis (they are S.7.2’s undeterminable holdouts), could raise raw for upstream mineral nodes.
- **Can return no.** After a few passes of §4 authoring settled at Pass S close, if raw outbounds stabilize below 2.0448 for several passes, the K.1 warning does not fire and FR-B holds as a genuine one-time anchor.

If the answer becomes yes: FR-B is tracking the max on a lagged schedule (the second anchor cycle), the K.1 warning was correct, and FR-C’s declared-arbitrariness would have been safer. That is the refuting outcome the original Q8 condition could not produce.

Adopted: the raw-outbound-above-2.0447548854281186 test replaces Q8’s drift-still-reports-4 test. Recorded in the ledger below.

5 — Restated recommendation

The corrections above change the arithmetic underlying the FR-B vs FR-C comparison:

crit ^{erion}	FR-B	FR-C
tier changes under derived boundaries	5	2
K.1 warning applies syntactically?	yes (fixed = max)	no (declared arbitrary)
K.1 warning applies on pattern reading?	possibly (item 4 test)	no
copper at concentration exactly 1.0?	yes	no (0.8180)
dynamic range restored at top?	yes	partially (top compressed)
ASML movement	crit → moderate (2 tiers)	crit → moderate (2 tiers)
AMAT movement	moderate → none	moderate → none
downstream tier movement (dysprosium/gallium/TSMC)	3 nodes move	0 nodes move
falsifiable flip test post-ship	yes (item 4 replacement)	not applicable (declared arbitrary)

Does FR-B still win once ASML’s true 2-tier movement is on the table?

No. The corrected picture makes the choice tighter than Pass T claimed:

- ASML’s movement is a wash — **both** candidates move ASML 2 tiers.
- AMAT’s movement is a wash — **both** candidates move AMAT 1 tier.
- FR-B additionally moves **dysprosium, gallium, and TSMC** — three more scored nodes changing tier, each of which the ship-pass would need to attribute to the fixed_reference change specifically (rather than to any node-level structural event).
- FR-B invites the K.1-warning discussion; FR-C sidesteps it by declaring the constant arbitrary rather than reading it off the graph.

The dynamic-range argument for FR-B (copper at exactly 1.0, ASML at 0.866, TSMC at 0.857) was a philosophical win, not a blast-radius win. The Pass T recommendation folded it into a claim that FR-B was preferable *overall*; the corrections invalidate the “overall” and leave the dynamic-range argument standing alone against a smaller blast radius and a cleaner relationship to the K.1 warning.

Revised recommendation: FR-C, fixed_reference = 2.5, boundaries derived at SF=3.0 (critical 0.5247316525037853, high 0.42320867926942163, moderate 0.15668443545638666).

Case against FR-C, stated fairly: the concentration axis is compressed at the top (copper at 0.8180 rather than 1.0; ASML at 0.7084; TSMC at 0.7010). The three clamped-under-FR-A nodes stop being tied at 1.0, but they are not distinguishable at the axis-max either. If the ship pass values “*copper is the outbound anchor, the concentration axis reads that clearly*” over “*minimize blast radius,*” FR-B is defensible on that ground alone — but the ship pass then owes the reader an explicit attribution for each of the 5 tier movements FR-B causes, particularly the 3 that FR-C would leave untouched.

FR-B remains a legitimate alternative if the ship pass author reads the dynamic-range argument as load-bearing. It is not the recommendation any more.

Ledger — Pass T addendum

- **ASML tier under FR-B derived is moderate, not high.** Pass T Q9 was correct; the ledger sentence “*they stay high*” was wrong. Retracted.
- **FR-B causes 5 tier changes under derived boundaries, not 2.** The ledger sentence “*Net movement: 2 nodes drop under FR-B derived versus 4 under the current drift comparison*” had the direction inverted and the counts wrong. Retracted.
- **applied_materials is a genuine mover.** Under both FR-B and FR-C, AMAT’s severity drops from 0.2015 to 0.1647 / 0.1347 respectively, and its tier drops from moderate to none under both candidates’ derived boundaries. Outbound-dominant in both states.
- **Pass T Q8’s flip condition was circular.** The K.2.2 §6.2 pattern’s fifth instance. Replaced with a falsifiable condition: “*after FR-B ships, does any future pass’s authoring push a node’s raw outbound above 2.0447548854281186?*” If yes, FR-B was tracking the max on a lagged schedule.
- **Revised recommendation: FR-C.** Smaller blast radius (2 vs 5), cleaner relationship to the K.1 warning, matches FR-B on ASML and AMAT movement. FR-B remains defensible on the dynamic-range argument alone; ship decision belongs to Weston.
- **Every claim in this addendum reads from pass_t_facts.json at commit 1dbdd46.** No scoring change, no data change, no config change —

the addendum is a reporting correction. Suite unchanged at 117 pass + 1
skipped + 0 xfail.